

Navigating Mental Health and Cannabis Use Post-cannabis Legalization: Experiences from Racialized Community Members

Contemporary Drug Problems

1–20

© The Author(s) 2025

Article reuse guidelines:

sagepub.com/journals-permissions

DOI: 10.1177/00914509251394526

journals.sagepub.com/home/cdx



Tanya Lazor^{1,2} , Kamna Mehra^{1,2}, Jennifer Rup^{1,2}, Jessica L. Wiese^{1,2} , Tara Marie Watson³, Hayley Hamilton^{1,2}, Sanjeev Sockalingam^{4,5,6,7}, Stefan Kloiber^{2,4,5,8}, Shari Margoese⁹, Samantha Wells^{1,2,5,10}, Akwasi Owusu-Bempah¹¹, Branka Agic^{10,12}, and Sergio Rueda^{1,2,4,5,13}

Abstract

Through conversations with members of racialized communities, we aimed to: explore perspectives shared by members of racialized communities regarding the relationship between cannabis and mental health; develop a greater understanding of racialized community members' mental health service interactions in relation to cannabis use and mental health; and examine whether various social identities interact to influence experiences with cannabis use and mental illness. From January to June 2022, we conducted semi-structured interviews with 26 members of racialized populations who were ≥18 years old, had used cannabis in the last 6 months, and had been in contact with the mental health sector across Ontario in the past year for a known psychiatric diagnosis. Many participants were 25–34 (46%), Canadian citizens (89%), and heterosexual (50%), with representation from ten different ethno-racial identities. Seventy-three percent of past-month cannabis users reported daily use. We identified five themes: experiences of the relationship between cannabis use and mental

¹CAMH, Institute for Mental Health Policy Research, Toronto, Canada

²CAMH, Campbell Family Mental Health Research Institute, Toronto, Canada

³Department of Family and Community Medicine, Temerty Faculty of Medicine, University of Toronto, Toronto, Canada

⁴Institute for Medical Sciences, University of Toronto, Toronto, Ontario, Canada

⁵Department of Psychiatry, Faculty of Medicine, University of Toronto, Toronto, Ontario, Canada

⁶Centre for Mental Health, University Health Network, Toronto, Ontario, Canada

⁷The Wilson Centre, University Health Network, University of Toronto, Toronto, Ontario, Canada

⁸Department of Pharmacology and Toxicology, University of Toronto, Toronto, Canada

⁹Independent Community Consultant, Toronto, Canada

¹⁰Dalla Lana School of Public Health, University of Toronto, Toronto, Canada

¹¹Department of Sociology, University of Toronto, Toronto, Ontario, Canada

¹²CAMH, Provincial System Support Program (PSSP), Toronto, Canada

¹³Institute of Health Policy, Management and Evaluation, University of Toronto, Canada

Corresponding Author:

Sergio Rueda, PhD, Senior Scientist, Institute for Mental Health Policy Research, Centre for Addiction and Mental Health, 250 College Street, Room 927, Toronto, Ontario M5T 1R8, Canada.

Email: ruedagento@gmail.com

health; cannabis use in response to barriers encountered with formal mental health supports; negative experiences with mental health services related to race, gender identity, and cannabis use; adverse effects of discrimination on mental health and cannabis use; and strategies to improve mental health programs. Interviews facilitated a deeper understanding of the complex relationship between cannabis use and mental health outcomes among racialized individuals. Clinical practice guidelines and training are recommended for healthcare providers to enhance culturally sensitive care regarding cannabis use and mental health. Research exploring risks and benefits of self-medication using cannabis would enrich our understanding of its implications specifically for this population.

Keywords

cannabis, legalization, mental health, substance use, qualitative research, ethnic and racial minorities

Introduction

Research demonstrates a high degree of comorbidity between mental illness and substance use disorders in Canada (Khan, 2017). Further, the relationship between cannabis use and mental illness is complex. The prevalence of weekly or more frequent cannabis use among individuals with mental illness is at least double that of the general population (Konefal et al., 2019), and rates of cannabis use disorder are also higher in people living with schizophrenia than in the general population (Hunt et al., 2018; McLoughlin et al., 2014). Some evidence suggests that cannabis use may increase the likelihood of developing mental illness. Cannabis use is estimated to increase the risk of developing psychosis by 40% among genetically predisposed individuals (Henquet et al., 2005). Several meta-analyses have demonstrated an increased likelihood of developing depression among those who use cannabis (Gobbi et al., 2019; Lev-Ran et al., 2014; Mammen et al., 2018; Smolkina et al., 2017). Other studies suggest that people who have a mental illness may use cannabis to alleviate symptoms (Andries et al., 2015; Bisaga et al., 2015; Hoch et al., 2019; Jetly et al., 2015); however, a systematic review of randomized controlled trials found mixed evidence on the effectiveness of cannabis for the treatment of mental health disorders (Hoch et al., 2019). When offered as adjunct to pharmacotherapy and psychotherapy, cannabis-based medicines were associated with improvements of several symptoms of mental disorders but not with remission. Further, mediating factors such as age of onset of cannabis use may influence the association between cannabis use and poor mental health outcomes (Day et al., 2015; Kiburi et al., 2021; Schubart et al., 2011). Overall, the extent to which cannabis use contributes to or alleviates mental illness remains unclear.

Nevertheless, emerging research shows that the use of cannabis as self-treatment to manage mental health symptoms is becoming increasingly common (Chou et al., 2019; Stuart-Maver, 2020) and daily cannabis use is increasing among those experiencing psychological distress (Weinberger et al., 2019). In the context of legalization and increased availability of high-potency cannabis products, researchers and healthcare providers (HCPs) are voicing concerns about the impacts of cannabis use on mental health and substance use needs as well as service uptake. Little is known about the intersection between mental health and cannabis use within this context, and this gap is particularly evident in research involving racialized populations.

Feedback from our study's Community Advisory Committee (CAC) indicated that the preferred term to use in reference to immigrant, refugee, ethno-cultural, and racialized individuals is "racialized." We have applied this term throughout this paper when referencing members from these diverse groups with the understanding that the term "racialized" is a societal construct that extends beyond categorization by race to the idea that members of these groups have been affected by racism or discrimination (Government of Ontario, 2024; Souissi, 2022). Research on cannabis use among racialized groups has

noted some trends in cannabis use. A 2025 study on past-month cannabis use among older adults identified notable increases among specific racialized groups, including non-Hispanic American Indian or Alaskan Native, Asian, Native Hawaiian or Other Pacific Islander, and multiracial individuals (rising from 1.6% in 2021 to 7.4% in 2023) (Han et al., 2025). Jeffers et al. also observed that high-frequency cannabis use was more prevalent among younger and racial minority populations (Jeffers et al., 2021). However, while both studies stratified cannabis use trends by demographic characteristics, neither provided an in-depth analysis of the contextual factors driving these patterns. Further, research in this area within the context of mental health outcomes is notably sparse. A 2019 report by the Mental Health Commission of Canada identified a significant lack of research focused on cannabis use among racialized populations (Clement et al., 2019) despite race, ethnicity, and immigration status being important determinants of health that affect mental health and substance use outcomes (Harrell et al., 2003; Utsey et al., 2008). Further, with the exception of some improvements made in recent years (Government of Canada, 2019, 2022, 2023), including Ontario's *Data Standards for the Identification and Monitoring of Systemic Racism* (Government of Ontario, 2024), uptake of routine and standardized collection of data concerning race and ethnicity in health administrative databases has been slow. While it has been 6 years since the 2018 federal legalization of cannabis in Canada (Government of Canada, 2021), it remains a crucial time to strengthen the evidence base and monitor the impacts of cannabis use on mental health outcomes within racialized populations.

Rates of mental illness vary across and within racialized populations, reflecting unique risk and protective factors. There is some evidence that racialized groups experience higher rates of mental illness and report lower mental health service utilization than the general population (Bansal et al., 2014; Chen & Kazanjian, 2005; Sentell et al., 2014). For example, some immigrants in Canada report using fewer mental health services (e.g., visits to physicians, psychiatrists, and hospitals) (Chen & Kazanjian, 2005), even among those experiencing severe mental illness (Bansal et al., 2014; Sentell et al., 2014). Different rates of treatment-seeking for mental health and substance use are associated with cultural differences in perceptions of mental illness (Ng, 1997), individual factors such as feelings of shame and stigma (Bradby et al., 2007; Leong & Lau, 2001; Loya et al., 2010), as well as barriers related to a number of other social determinants of health. Researchers and advocates have also noted the lack of culturally appropriate mental health services and supports for diverse populations (Boksa et al., 2015; Kafele, 2004).

Understanding the link between cannabis use and mental health among racialized populations requires consideration of the history of drug laws and how they negatively impact racialized communities. Historical analyses indicate that Canada's earliest drug laws, including cannabis laws, were largely intended to control racialized communities (Carstairs, 2006; Giffen et al., 1991). For many years, profiling and over-policing of racialized communities have resulted in the unequal enforcement of drug laws contributing to disproportionate rates of charges and convictions (Khenti, 2014; Marshall, 2015; Mosher & Akins, 2015; Owusu-Bempah & Gabbidon, 2020; Owusu-Bempah & Rehmatullah, 2023; Tanovich, 2006; Wortley & Tanner, 2003). Racism and criminalization affect the physical and mental health of members of these communities (Harrell et al., 2003; Utsey et al., 2008) and experiences of poverty, marginalization, and other social determinants of health also impact mental health and substance use (Seng et al., 2012). Experiences with oppression, social exclusion, poverty, and minority stress have been found to uniquely predict motivations for cannabis and other substance use among racial and gender minorities (Buckner et al., 2016; Scheim et al., 2017). Further, there are important historical, religious and spiritual perspectives related to both cannabis and mental health that should be considered within their cultural or social contexts in research within racialized populations.

Overall, understanding the experiences, needs, and motivations of different racialized populations in relation to the intersection between mental health, cannabis use, and mental health service use is essential to promote greater representation and inform meaningful service improvements and culturally

appropriate care. More research is needed to address these gaps in understanding of the racial and ethnic differences in cannabis use, including understanding attitudes toward use and further elucidating reasons for consuming. Given the unique mental health challenges faced by racialized populations and recognizing intersecting identities, research is also needed to better understand the potential therapeutic benefits of cannabis to manage symptoms of mental illness, especially within unique cultural contexts. Finally, it is also essential to explore participant perceptions of the mental health sector related to the intersection between mental illness and cannabis use.

To promote and enhance health equity, an intersectional lens needs to be applied in this work. Under-represented populations are highly diverse and inclusive of many intersectional social identities related to race, class, gender, sexual identity, and other social determinants of health.

Thus, we used a strengths-based approach to examine the experiences and perspectives of these understudied priority groups, with the goal of advancing knowledge regarding cannabis and mental health. Specifically, we aimed to: (1) explore key experiences and perspectives shared by members of racialized populations regarding the relationship between cannabis and mental health; (2) develop a greater understanding of racialized community members' mental health service interactions and utilization in relation to cannabis use and mental health; and (3) examine whether and how various social identities interact to influence perspectives and experiences with cannabis use and mental illness.

Materials and Methods

Theoretical Approach

As noted above, cannabis use in the context of mental illness experienced by racialized populations is uniquely experienced and compounded by the intersectionality of various inequities related to race, gender, ethnicity, sexual identity, socioeconomic status, and other social determinants of health. As such, we applied a lens of intersectionality as a theoretical framework for researching health inequities among racialized populations with mental illness who use cannabis. Intersectionality as a theoretical framework has been applied in the field of public health research (Bowleg, 2012; Heard et al., 2020; Macgregor et al., 2023) and highlights the ways that systematic barriers faced by under-represented populations are exacerbated by the interconnected systems of power, thereby perpetuating health inequities (Macgregor et al., 2023). By applying a lens of intersectionality, we aimed to center participants' lived experiences and better situate them within the social context.

Setting

We conducted a qualitative study with members of racialized populations who are living with mental illness, use cannabis, and are in contact with the mental health sector across Ontario, Canada. We recruited participants from hospital treatment programs and community services and through contacts with partners across the province to reach community agencies and remote locations. We primarily recruited through three key hospital hubs, programs, and clinics: (1) The Mood and Anxiety Ambulatory Services (MAAS) and General Assessment Clinic (GAC) that receives referrals from the community and internally; (2) the ECHO Ontario Mental Health (ECHO-ONMH) at CAMH and the University of Toronto (a provincial tele-education hub-and-spoke mental health and substance use capacity building effort in Ontario) (Sackalingam et al., 2018; Zhou et al., 2016). We also recruited through the CAMH Research Registry, which contains information of individuals that are interested in participating in research studies.

Community Advisory Committee

To describe and give voice to the lived experiences of our population of interest, we established a CAC with representation from organizations and people with lived experience from racialized communities. The CAC participated in development of study materials such as the interview guide, analysis of the interview data, and knowledge dissemination. We recruited ten members through CAMH partners and networks to ensure sufficient stakeholder representation of intersecting identities. We emphasized equity and shared decision-making throughout the research process.

Recruitment and Data Collection

Interviews were conducted from January to June 2022. Eligible participants were at least 18 years of age, identified as members of racialized communities (e.g., Black, East Asian, Southeast Asian, Indigenous (First Nations, Métis, Inuk/Inuit), Latino, Middle Eastern, South Asian), had used cannabis in the last 6 months, had accessed mental health/substance use services for a known psychiatric diagnosis in the past year, and were able to complete an interview in English. We used purposive sampling and focused on the primary client base of CAMH and its Ontario-wide partners. Recruitment posters were shared with the assistance of clinical staff and our CAC members, and via social media. When potential participants contacted the study coordinator, they were provided with a study description and completed an eligibility assessment. Of the 32 people assessed for eligibility, one was not eligible as they did not identify as “racialized,” and one was not eligible because they had not accessed services for mental health concerns. The remaining 30 individuals were eligible, of which 29 consented to participate in the study and 26 completed a qualitative interview.

Data were collected via semi-structured one-to-one interviews. Due to the COVID-19 pandemic, all interviews were conducted over the telephone. Interview length averaged 56 minutes and ranged from 26 to 93 minutes. The interviewer was not known to participants and had previous experience conducting qualitative interviews. All participants completed an informed consent discussion with the study coordinator and provided electronic informed consent prior to the telephone interview. The interviews were audio recorded and an external service transcribed the interviews verbatim and anonymized all identifying information. Participants were compensated with a \$30 electronic gift card. The study was approved by the CAMH research ethics board (#055/2021) and all procedures were performed in compliance with relevant laws and institutional guidelines.

Interview Guide

The study team and CAC members co-developed the demographic survey and interview guide. Demographic information was collected by phone before the interview was recorded, including town/city of residence, age, gender, sexual orientation, ethno-racial identity (Government of Ontario, 2024), immigration status, level of education, mental health services accessed in the past 12-months, last cannabis use, past 12-month cannabis use, reason for cannabis use, whether they had a medical document authorizing the use of cannabis for medical purposes, and modes of cannabis consumption.

Participants were asked about their attitudes and perceptions of cannabis use, and how consumption may be related to their experiences with mental illness and intersectionalities such as race and gender identity. They were asked to reflect on their contextualized experiences interacting with the mental health and substance use sector in relation to their mental health and cannabis use, and experiences of discrimination and racism. Participants were also asked to reflect on their experiences with therapeutic benefits and negative impacts of cannabis use, to consider their main motivations for cannabis use, and how motivations may have changed over time or be related to experiences connected to their mental health and identity.

Analysis

The demographic survey data were analyzed descriptively (e.g., means, frequencies) using SPSS statistical software (IBM Corp., 2017). Audio files from the interviews were transcribed and analyzed using NVivo 12.0 qualitative analysis software (Ltd, 2020) in conjunction with data collection in an ongoing, iterative process. Analytic notes and memos were taken during the interviews, while coding, and during the development of themes. One team member completed all interviews and coding. A codebook was developed using thematic analysis as described by Braun and Clarke (Braun & Clarke, 2006), which describes six steps of thematic analysis: (1) familiarize; (2) generate codes; (3) generate themes; (4) review themes; (5) determine theme significance; and (6) report findings. One team member conducted open coding of five interview transcripts and their corresponding memos. The five interviews were selected based on the richness of the data provided by the participants to ensure the initial codebook would have as many codes as possible. Members of the study team discussed the codes after the first five interviews and the codebook was revised based on feedback. The remaining 21 interviews were coded after the codebook was edited, however, developing the codebook was an iterative process, and new codes were added as they were identified while coding the remaining interviews. Regular team meetings with in-depth discussions about emerging themes also helped ensure data elements were not overlooked, and a broad lens was utilized during analysis.

Reflexivity

By applying an intersectionality framework, the research team recognized the importance of reflexivity and acknowledging their own social identities within the context of their role in the research process. For example, the interviewer and coder identified as an immigrant and racialized heterosexual woman and considered the implications of her own social identities, experiences, biases, and privileges of belonging to this position in the research process.

Results

A total of 26 people participated in the interviews. Participant demographic characteristics are shown in Table 1. Participants represented a variety of ethno-racial identities including Asian (East Asian, South Asian and Southeast Asian), Black (Black African, Afro-Caribbean and African Canadian), First Nations, Indo-Caribbean, Latino, and mixed race. Half of the participants were women and half identified as heterosexual. Table 2 displays participant cannabis use trends. Ninety six percent of participants had used cannabis within the past 30 days and 73 percent reported daily use. Most participants cited consuming cannabis for recreational or medicinal purposes (65 and 81 percent, respectively).

Through thematic analysis, five themes were identified: (1) Experiences of the relationship between cannabis use and mental health; (2) Cannabis use in response to barriers encountered with formal mental health supports; (3) Negative experiences with mental health services related to race, gender identity, and cannabis use; (4) Adverse effects of discrimination on mental health and cannabis use; and (5) Strategies to improve mental health programs.

1. Experiences of the relationship between cannabis use and mental health:

Navigating Positive and Negative Outcomes of Cannabis as a First-Line Defense for Mental Health Management

Woven throughout many of the discussions with participants were the perceived impacts of cannabis use on their mental health conditions and associated symptoms. Participants spoke about several

Table 1. Demographics of the Study Cohort (N = 26).

Variable	N (%)
Age	
18–24	6 (23%)
25–34	12 (46%)
35 +	8 (31%)
Gender identity	
Woman	13 (50%)
Man	10 (39%)
Non-binary or gender fluid	3 (11%)
Sexual orientation ^a	
Heterosexual	13 (50%)
Bisexual	5 (19%)
Gay	3 (12%)
Pansexual	3 (12%)
Queer	3 (12%)
Ethno-racial identity ^a	
Black	9 (35%)
East Asian	5 (19%)
Southeast Asian	4 (15%)
South Asian	4 (15%)
Other ^b	6 (23%)
Immigration status	
Canadian Citizen	23 (89%)
Landed/Permanent Resident/International Student	3 (12%)
Level of education	
Less than high school	2 (8%)
Completed high school	5 (19%)
College, CEGEP or other non-university certificate or diploma	4 (15%)
University/Post-graduate	15 (58%)

^aEthno-racial categories align with Ontario's Data Standards for the Identification and Monitoring of Systemic Racism (Government of Ontario, 2024) race categories; also in alignment with these standards, participants had the option to select more than one category making the total sum >26.

^bDue to a small number of participants for some ethno-racial identities (Indigenous, Latino, Middle Eastern, Mixed race), these identities were combined into "other" to reduce concerns with identification.

positive and negative outcomes related to their use. Many perceived positive outcomes were in relation to helping to create a sense of calm, improving mood, and assisting with sleep:

If I'm not feeling great or I'm feeling, like, super anxious, a lot of time, cannabis is my first line of defense, because again, it's like, so accessible. So I might like smoke a little bit, and then start to meditate or then go for my walk, or, you know, then try to do some of my other strategies. So it's almost like a tool in the tool kit.

In contrast, some perceived negative mental health outcomes described by participants included worsening anxiety and a general sense of a negative mental state:

If I'm already in a bad mental state, before I take it, I don't feel great after it. I start thinking of like, very, I don't know, like, philosophical scary things. Like, you know, what the meaning of life is, and I start freaking out. So [...] sometimes works with my anxiety and makes it worse. And I guess it all depends on, like, the edibles, for

Table 2. Cannabis Use Among Study Participants (N = 26).

Variable	N (%)
Past 12-month cannabis use	
Daily	19 (73%)
Weekly	2 (8%)
Monthly	3 (12%)
Less than monthly	2 (8%)
Reasons for cannabis use ^a	
Recreational	17 (65%)
Medicinal/therapeutic	21 (81%)
Spiritual/Other	6 (23%)
Medical document for cannabis	
Yes	4 (15%)
No	22 (85%)
Mode of consumption ^a	
Eating	23 (88%)
Drinking	9 (35%)
Smoking	25 (96%)
Oral capsules	5 (19%)
Vaping	20 (77%)
Mist/Other	3 (12%)
Topically	6 (23%)

^aParticipants had the option to select more than one category.

sure make my anxiety worse. But the oils and the vaping really help. So, increased anxiety is actually a negative aspect. Sometimes, very, like, vivid dreams are the negative aspect.

Participants also described both positive and negative physical outcomes such as increased appetite, coughing, and dependency:

It's also the only thing that is helping me be able to eat.

Negative effects I would say, sometimes, it can even be physical, like, just the coughing. And [...] I would even say there are periods of dependency that I feel I experience.

A few participants also described symptoms that had both a positive and negative aspect to them. This was often in relation to consuming too much or a high-potency product. One spoke about the close relationship between the strength of the strain they use and the ability for cannabis to either calm or trigger symptoms of anxiety:

THC can also trigger some of my symptoms, which is anxiety. So that's why it's imperative for me that even if I am smoking a product or using a product that has, say, twenty five percent THC, I have to make sure that there is higher CBD counterbalance. Right? Or else, it'd be counterproductive to keep using something that's just going to worsen my symptoms.

For the most part, I'd say that it does help to relax, destress. But I do think that sometimes, I don't know why, maybe if I consume too much, [...] mental health, sort of symptoms will come back, so probably feelings of anxiety will come back, things of that sort.

Others spoke about feelings of dissociation which they described as helpful at times and not ideal at other times:

I think emotionally, it's helped - it's created some dissociation, which can be, I think, good and bad. And the bad situation of it is that if there is a feeling that I'm experiencing or a perspective that I'm engaging with or some conflict that I'm in the midst of, and I use cannabis to try to subside it, I think sometimes that can prevent, that can create a level of dissociation and a level of stepping away from emotion [...]

Participants reflected on the nuanced ways in which they use cannabis as a tool to manage their mental health. Generally, participants reported a range of experiences—positive, negative, or both—related to their cannabis use. They demonstrated significant insight and self-awareness regarding the difficulties navigating mental health challenges, and the aspects of cannabis use that they found to be problematic. Participants acknowledged the complexities, benefits and risks of using cannabis as a coping tool, along with strong familiarity with the effects that cannabis use can have on them. Participants largely spoke about using cannabis with an intention to address a specific symptom, and this practice was generally reported to be helpful for many aspects of their experience with mental health symptoms. Participants demonstrated strategic decision-making regarding their cannabis use, resilience, and dedication to optimizing their wellness by balancing the benefits and drawbacks when possible.

2. Cannabis use in response to barriers encountered with formal mental health supports:

Turning to Cannabis When Systems Fall Short

Participants spoke at length about various barriers to formal mental health supports and how these barriers related to their cannabis use. The stigma surrounding mental health was apparent in certain cultures and communities, particularly when participants spoke about the reactions their family members had to the idea of therapy or using medications for mental health. For some, the fear of disobeying or going against cultural norms may have prevented them from seeking treatment:

Mental health is stigmatized in my family. They assume, 'Oh, you're weak.' [...] I think that in the Black community in general, you think 'Oh, that's like a White man's condition, to have these things.' They don't understand like, these are universal issues, that affect all races. —

The high cost of formal mental health supports were described as another barrier to seeking treatment, as well as lack of access to mental health services and supports due to long wait times, difficulties connecting with a HCP, and lack of knowledge about available resources:

Access needs to be prioritized... like, if you don't google, if you don't research, if you don't, it's not well known what is out there available, what is covered by [the Ontario Health Insurance Plan]... if there are these programs out there, go to the communities that need them and announce it, like make it be known. The majority of the resources that I found, I had to search for them [...]. Not everybody is going to do that.

To mitigate these barriers, many participants spoke about consuming cannabis to manage or support their mental health instead of accessing formal mental health supports. Cannabis use was described as an easier way of coping with negative feelings and distress compared to therapy. This was due to factors such as ease of access to cannabis and the immediate relief cannabis provided:

[Cannabis is] readily available, versus you have to book an appointment and you have to commute there, and oftentimes, they can't see you right away, so you're like, waiting in reception to be seen. And then when you go in there, like, you don't get enough time, one on one, with the doctor or the support worker. [...] And then, like once you leave, [...] you go right back to your same state of like, confusion and anxiety, and depression.

Some participants spoke about the benefit of cannabis as an immediate but not long-term support, while others spoke about feeling unheard by their HCPs who would encourage psychiatric medications or not engage in discussions about cannabis:

So I kept going to [my family doctor] and asking him [about cannabis] and he didn't want to hear it. 'You have anxiety, fine. Take Wellbutrin.

This theme reflected the ways participants used cannabis as a pragmatic and accessible alternative when formal mental health supports were inaccessible, were culturally stigmatized, or did not meet their needs. Many participants explained that the barriers they had encountered with formalized mental health supports could be addressed, at least in part, by cannabis. However, some noted that cannabis alone was not effective in managing mental health symptoms, while others noted that collaborating with HCPs on a treatment plan that included cannabis was difficult. Participants demonstrated resourcefulness in navigating various cultural and systemic barriers and reinforced the need for culturally safe and accessible care.

3. Negative experiences with mental health services related to race, gender identity, and cannabis use:

Navigating Conditional Trust and Identity-Based Harm in Mental Health Services

Many participants felt that their race affected how they were treated when they accessed mental health services, sometimes when the HCP was of a different race than the participant. Participants eluded to potential unconscious biases and stereotypes held by HCPs which created discomfort, negative feelings, and a lack of confidence in the treatment process. One participant spoke about microaggressions they faced when interacting with HCPs:

Yeah, they just assume you're stupid, cause you're a Black person. You get a lot of prejudice, no matter who you're dealing with. [...] You get a lot of stereotyping... people try to finish your sentences for you... you feel like they're not really listening to you. They're talking at you, but not to you. So, I feel like I'm just wasting my time.

Some participants noticed a positive change regarding the current approach to addressing or acknowledging gender identity in healthcare, compared to treatment in the past. However, others felt that there were still general feelings of discrimination or judgment of the experiences of those who identify as gay or queer, and noticed a lack of knowledge or judgment towards different gender identities:

Being sexually active and being queer, people stigmatize me for having sex or having unprotected sex [...]. I've experienced some judgement actually [...]. I'm afraid that people either won't take me seriously or they won't diagnose me in a way that is differential.

Participants spoke about how preconceived expectations and stereotypes about their race and genders affected the response to their healthcare needs. Women in particular spoke about not being taken seriously, an experience that was heightened when interacting with HCPs:

I do think it's Black women are not believed in medical settings. I mean, that is a fact of where, when we complain about pain, we're perceived as like, being dramatic and stuff. So there have been times when I just feel like, as a Black woman, when I'm talking about my experiences, how I feel with my levels of depression and stuff like that, I always, I haven't always been listened to, instead I've just been thrown a prescription.

Some participants described negative experiences with mental health services related to race and gender identity within the context of their cannabis use. The need for comfort and trust with a HCP to be able to disclose use was identified as important:

Definitely in the first couple of years of my use, I didn't tell anyone, for sure. And then, you know, mid-way through my substance use journey, I did try to share with my family doctor, and again, like I said, I didn't like their response. And so I kind of disengaged from primary care.

Many participants also reported feeling that HCPs often focused on only one area of their identity rather than considering how all of their intersecting identities influenced their mental health. Not addressing the myriad layers of participants' experiences and social identities reportedly compounded their mental health concerns. Some participants expressed feeling that it was hard for HCPs to understand the complexity of their experiences:

I find that white LGBTQ people don't understand what it feels like to have to identify as your race plus your gender identity or your sexuality and you don't understand like the intersectionality of these identities.

One participant conversely described compartmentalization of their intersecting identities in such a way that it played less of a role in their cannabis use or experience with HCPs:

In terms of intersectionality, I don't think that, I guess, has much of a role in terms of cannabis usage or my, or mental health. [...] I kind of compartmentalize them and kind of treat them separately, at least in terms of healthcare and mental health interactions. [...] I think there is probably, like, some degree of intersectionality, but I don't think it affects, really, my cannabis usage or my, yeah, my, or anything related to healthcare. But yeah, I guess like I do prefer them to kind of be treated separately.

This theme reflected the ways in which participants' trust in care providers has the potential to erode due to intersecting systems of oppression and highlights the emotional toll that navigating access to these supports can require of members of racialized populations. Many participants described negative interactions with HCPs that were often related to participants' race, gender, identity, cannabis use or other social identities and related systems of discrimination. These negative interactions were said to impact patient engagement with their provider and providers' responses to patient needs. Specifically, participants depicted a perceived lack of therapeutic safety that directly impacted care, including having to consider the risk associated with disclosing their cannabis use. Participant's responses articulated the emotional complexity and discernment required of them to engage with these services.

4. Associations between adverse effects of discrimination on mental health and cannabis use:

Cannabis use in the Face of Racialized Mental Health Experiences

For some participants, experiences with race and gender discrimination adversely affected mental health by contributing to feelings of worthlessness and a lack of a sense of belonging in society. Participants also noted that discrimination led to a lack of trust in HCPs, as well as feelings of doubt about the validity of their mental health concerns:

I've had, like, a lot of different experiences with racism and it definitely affected my mental health very negatively, because I would experience feelings of worthlessness, so when I'm experiencing racism, it kind of makes those feelings of worthlessness kind of just surge. Because I feel that people think that I'm worthless, just because of my race, or think that I'm not good enough because of my race. And growing up in a predominantly white area, it was pretty prominent to receive different comments about my race.

For some participants, cannabis was used as a form of coping with discrimination, as it provided relaxation, relief, or distraction from negative feelings:

The existentialism of being discriminated against, I think I am always going to be like, I still have to deal with it on my own. And I feel like cannabis use relaxes me enough to be grounded in the decisions and the choices that I have or will make, when it comes to my body.

Other participants also described cultural, societal, and religious stigma against cannabis as factors impacting their cannabis use:

I think there's a race component. I think if you're white, it's seen as just a hobby, as a recreational, you know, activity. So, we even saw, like, with legalization, now there's – I don't know if you've seen it but like, every other store in [city] is a cannabis shop. (...) So for white Canadians, it's equivalent to visiting the [liquor store]. Their spending habits are not questioned, or their morals and values are not judged. But when it's people of color, like Black people or Indigenous people, we're seen as like bums, wasting our money on drugs when, you know, it's like, statements are made, like 'Oh, they're probably on welfare and look where they're spending their money.' or 'They don't own a house, and they're spending their money on this. Why don't they get a job?'

This theme articulated participants' use of cannabis as a self-directed coping strategy in the absence of formal supports, while also navigating the compounding stigma associated with its use. Participants highlighted the impact that social minority group membership can have on mental health and experiences as a consumer of cannabis. Some individuals described a double-standard in public discourse in which cannabis use by members of majority groups is normalized, while in members of racialized groups it is morally judged, leading to experiences of shame for some participants. These chronic and cumulative experiences have the potential to deeply impact mental health and overall sense of belonging, and also underscore the emotional resilience and critical awareness that racialized individuals have as a result of navigating these nuanced settings.

5. Strategies to improve mental health programs:

Reimagining Mental Health Services: Centering Cultural Relevance, Representation, and Equitable Access

Participants spoke about several aspects of mental health programs and services that require improvement. There was a need for more acknowledgement of racialized experiences and systemic issues, as well as more diversity and representation of different races and identities in the healthcare sector at all levels, including supporting staff. Participants also noted a lack of inclusive mental health resources offered in languages other than English:

I think, also like, real in depth cultural-based trainings, for all support staff, the surrounding people, so it's not just the therapist themselves, like, cause it's like, the person's receptionist, that plays a role. [...] I know for some loved ones, language services has been a big thing, being able to access mental health services, but not in English.

One participant mentioned that the strategies and solutions offered in therapy are not always applicable to all cultural backgrounds, and there was a need for more HCP training and education to emphasize culturally sensitive approaches rather than a “one size fits all” approach:

When I say that I'm Asian that I'm from where I'm from, they would be like, 'Oh, now I understand like, you as an Asian woman from that background, you have to get married earlier. You have to have children, and you live in a really oppressed system. So, here's your set of struggles. I get that.' And, but even if they say they get it, they provide, like, unrealistic solutions or no solutions at all, because they would just be like 'Oh, like, you should talk to your family about this.' But I'm like, that's not, that's just not (laugh) culturally sensitive... they don't understand my cultural background, and because my culture is so different from theirs, a lot of, I guess, what they're taught to do, and a lot their strategies are not really applicable to people from other cultural backgrounds.

Conversely, some participants felt that it is unnecessary for services to emphasize race and cater towards cultural differences. Some participants felt that more focus should be placed on access to services, such as those for mental health and cannabis, instead:

Perhaps, if there was, again, not geared towards culture, but just an attempt [at] equal distribution of the services... throughout areas that everybody needs to the supports that anybody who does need to access them, can access them and there's absolutely no barriers.

Another suggested alternatives to accessing mental health programs, such as having safe and comfortable social spaces to consume cannabis:

I think even with cannabis, that there should be like, cannabis lounges that are safe consumption sites, in the same way that we experience clubs and bars, or maybe something that was like, less medically isolating, as I've witnessed other, maybe more severe safe consumption sites.

One participant suggested that offering more mental health programs in different formats that catered to the needs of specific communities, including online, in-person, or group therapy for people of color would be helpful:

Definitely group therapy for people of colour to talk would be great, different programs, specifically for children of colour and youth of colour, to be in a place where somebody can look like them and understand the experiences that they've been going through and definitely these groups should be led by a person of colour as well, just because there's more understanding within the community of people of colour, when it comes to mental health and just identity and everything. There's just really a bigger understanding than if it was coming from a non-person of colour.

Participants had a broad range of rich feedback related to potential improvements to enhance their experiences engaging with mental health supports and incorporating cannabis use into the spectrum of care. This theme captured participants' call for reforms to formal mental health supports to enhance inclusivity, culturally responsiveness and accessibility for racialized groups, with the overarching goal of dismantling barriers to effective mental health support.

Discussion

Cannabis use was commonly described by our study participants as an alternative to formal supports, as it was easy to access and provided immediate relief of mental health symptoms. Previous research

exploring cannabis use among populations facing mental health difficulties has noted similar themes. A study by Turna et al. administered an online survey to examine recreational and medicinal cannabis use patterns among adults in Ontario and found that nearly two thirds of participants who reported consuming cannabis for medical purposes used it as a substitute for alternative medications to manage symptoms related to pain, anxiety, arthritis, and to promote sleep and relaxation (Turna et al., 2020). A qualitative study of perceptions, experiences and patterns of cannabis use in individuals with mood and anxiety disorders by Das et al. similarly noted that nearly 40% of participants described consuming cannabis as a means of managing their mental health symptoms (Das et al., 2024). We similarly observed that participants often reported consuming cannabis instead of medication or other formal supports for symptom management. Our findings provide additional context to what was previously known by teasing out the specific experiences of racialized populations, allowing us to identify cultural stigma from friends and family members, as well as negative experiences with mental health supports, as common themes underpinning these decisions. In addition, our research sheds light on the practice of self-directed symptom management using cannabis with a unique focus on racialized populations and their reasons for avoiding engagement with more formalized supports. These findings suggest that more educational health information is needed for racialized individuals considering cannabis use as a means of mental health symptom management so that individuals have a thorough understanding of the effects of cannabis use in relation to mental health. This information might include the potential effects of the various products available within Canada's legal market, different routes of administration, and range of potencies.

Consistent with prior work highlighting the negative effects of cannabis use on mental health (Crippa et al., 2009; Degenhardt et al., 2003), some participants in our study perceived that cannabis worsened some of their symptoms, and some also noted that cannabis is not a suitable long-term treatment option. A cross-sectional study of medical cannabis users noted that only one quarter to one third of participants who reported consuming cannabis for symptom management also reported a reduction in the symptom they were aiming to alleviate (i.e., a reduction in anxiety when cannabis is reportedly used to manage anxiety) (Bonn-Miller et al., 2014). In contrast, other participants in our study perceived that cannabis was an effective alternative to formal mental health supports for symptom management. While self-reported cannabis use for symptom management is commonly described in the literature, it is possible that medical cannabis users in racialized populations may similarly only be gaining minimal benefit at most in terms of symptom relief for mental health struggles, or that people may misperceive that cannabis is helpful to them. However, racialized populations face additional barriers to accessing alternative, more formalized supports for mental health symptom management. Thus, a more thorough understanding of the immediate and long-term outcomes of cannabis use for management of mental health symptoms among racialized populations is critical to enhance symptom management and educate and engage HCPs. Further, a more thorough understanding of the relationship between intentions behind use and actual immediate and long-term outcomes of use is needed.

Our findings highlighted that discrimination negatively impacted attitudes towards health services for members of racialized populations who also belong to a gender minority population. Das et al. noted that participants reported limited involvement of HCPs when making decisions related to their cannabis use, and instead obtained information on cannabis products from non-medical sources such as websites and staff at cannabis retail stores (Das et al., 2024). These findings aligned well with perceptions reported by our study participants, who commonly identified a need for intersectional approaches in mental health programs and accessibility of cannabis, as well as more diversity, representation, and culturally sensitive training for HCPs. Others have called for sensitivity training for all individuals working with transgender people, including HCPs, to improve the quality of care provided (Winter et al., 2016).

The concept of training was further explored by Mukhopadhyay et al., who conducted a scoping review of educational interventions addressing systemic racism in post-graduate medical training (Mukhopadhyay et al., 2024). The authors identified a range of educational interventions that are

paving the way for implementation of holistic and sustainable approaches to anti-racist education for professionals in medical training (Mukhopadhyay et al., 2024). The interventions described by the authors mirror many of the improvements suggested by our study participants, including in-depth cultural-based training for all patient-facing staff and expanding access to and format of mental health programs. These findings contribute to a growing body of research highlighting the need for significant mental health service improvements including ensuring they are culturally appropriate. Many participants in our study described their experiences of discrimination within, lack of access to, and high cost of mental health supports as contributors leading to cannabis use as a form of coping and symptom management. Therefore, access to affordable and culturally sensitive mental health supports may potentially reduce people's tendency to use cannabis in place of more formalized supports.

Finally, many of our participants reported experiencing discrimination, or that HCPs only focused on one area of their identity rather than considering how all of their intersecting identities play a role in their mental health experience and cannabis use. These findings align with other research exploring the relationship between intersecting social identities and cannabis use. A recent systematic review by DeSon and Andover identified five studies which confirmed a positive association between sexual orientation-based microaggressions and cannabis use and problems (DeSon & Andover, 2024). A longitudinal cohort study by Dyar et al. noted that when sexual and gender minority individuals experienced more microaggressions or general stress than usual, they also experienced more cannabis use problems (Dyar et al., 2019). A similar study of sexual and gender minority individuals assigned female at birth were more likely to use cannabis if they experienced more microaggressions or victimization than usual (Dyar et al., 2020). Combined, these findings highlight the importance of HCPs having an awareness of each individual's unique set of social identities, mental health diagnoses and symptoms, and the ways that these interact with each other and impact a person's unique experience. It is critical that individuals feel comfortable consulting a HCP who can apply an intersectional lens to provide tailored consultation and guidance on an individual level.

Our findings make an important contribution to this growing body of research, as we were able to focus specifically on racialized populations. We learned that the decision to use cannabis in the context of a mental health diagnosis among this unique group is influenced by several additional factors related to cultural stigma, discrimination, and unconscious biases within the healthcare system. Our strengths-based approach centered participant voices and provided them with a safe space and the agency to speak about their lived experiences. The qualitative methodology used for our study allowed us to generate a rich, more nuanced understanding of the unique barriers faced by racialized populations interacting with the mental health and substance use sector in relation to their mental health and cannabis use.

Our findings must be considered in the context of the study's limitations. Eligible individuals who were approached for inclusion and chose not to participate may have had unique and differing opinions to offer that were not captured here. Second, it is likely that several eligible members of our study population were not aware of the study due to various barriers limiting access to care which may have been exacerbated by COVID-19 restrictions that remained in place at our organization at the time of the study. To mitigate this limitation, we attempted to extend the reach of our recruitment efforts through a variety of means and access points (i.e., by engaging Ontario-wide partners) and a variety of means (i.e., posters, word of mouth, social media). Finally, we were unable to distinguish whether participants' recollections of cannabis use were related to mental health or other purposes, which should be considered when interpreting the findings.

Conclusions

Participant interviews illuminated a complex interplay between cannabis use, mental health outcomes, and intersectionality of cultural, immigrant, racial, gender, and sexual identities. Our data analysis yielded five themes describing barriers to care in the context of differential treatment and cultural

sensitivity and the use of cannabis for mental health symptom relief. Our findings have several implications. They highlight the importance of prioritizing cultural sensitivity to and awareness of the intersectionality of identities of racialized individuals as a means of improving engagement with HCPs and reducing cannabis use as an alternative. They also illuminate the importance of engaging with members of racialized communities to further understand their experiences and perspectives, to inform mental health service improvements, and promote culturally appropriate care. Future research is needed to develop recommendations for healthcare service improvements as well as clinical practice guidelines for HCPs focused on provision of culturally sensitive care specifically for racialized populations who experience mental health challenges and use cannabis. Exploration of the potential implications of self-medication with cannabis for those with mental health concerns would also enrich our understanding of risks and benefits of cannabis use in this population.

Acknowledgements

We would like to thank the clinical staff members at CAMH and our Ontario-wide partners for their assistance with recruitment, as well as members of the CAC. Their contributions to study and interview guide design and recruitment was invaluable. We would also like to express our sincere gratitude for the valuable information provided by each of our study participants.

ORCID iDs

Tanya Lazor  <https://orcid.org/0000-0002-1852-7009>

Jessica L. Wiese  <https://orcid.org/0000-0003-3740-3607>

Ethical Approval Statement

The study was approved by the CAMH Research Ethics Board (#055/2021).

Informed Consent Statement

All participants completed an informed consent discussion with the study coordinator and provided electronic informed consent prior to the telephone interview.

Author Contributions

HH, SS, SK AO-B, BA, SM, TMW, SW, JLW, and SR contributed to study conceptualization, design of the methodology and funding acquisition. KM, JR, JLW, TL, and SR contributed to data collection, analysis and interpretation. SR provided study oversight, and JLW and KM contributed to project administration. TL and KM wrote the first draft of the manuscript, and all authors critically revised previous versions of the manuscript for intellectual content. All authors read and approved the final manuscript and agree to be accountable for all aspects of the work.

Funding

The authors disclosed receipt of the following financial support for the research, authorship, and/or publication of this article: This research was funded by a CIHR Catalyst Grant sponsored by the Mental Health Commission of Canada. The work of the Mental Health Commission of Canada is supported by a grant from Health Canada. The views represented herein solely represent the views of the research team and do not represent the views of the Mental Health Commission of Canada and Health Canada. SR was supported by an OHTN Endgame Innovator Award. TL is supported by a CANFAR/CTN + Postdoctoral Fellowship Award and a CIHR fellowship through the HIV/AIDS and STBBI Research Initiative (202210HIV-494755-255928).

Declaration of Conflicting Interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Data Availability

Due to the sensitive nature of the questions asked in this study, survey respondents were assured raw data would remain confidential and would not be shared.

References

- Andries, A., Frystyk, J., Flyvbjerg, A., & Stoving, R. K. (2015). Changes in IGF-I, urinary free cortisol and adipokines during dronabinol therapy in anorexia nervosa: Results from a randomised, controlled trial. *Growth Hormone & IGF Research*, 25(5), 247–252. <https://doi.org/10.1016/j.ghir.2015.07.006>
- Bansal, N., Bhopal, R., Netto, G., Lyons, D., Steiner, M. F., & Sashidharan, S. P. (2014). Disparate patterns of hospitalisation reflect unmet needs and persistent ethnic inequalities in mental health care: The Scottish health and ethnicity linkage study. *Ethnicity & Health*, 19(2), 217–239. <https://doi.org/10.1080/13557858.2013.814764>
- Bisaga, A., Sullivan, M. A., Glass, A., Mishlen, K., Pavlicova, M., Haney, M., Raby, W. N., Levin, F. R., Carpenter, K. M., Mariani, J. J., & Nunes, E. V. (2015). The effects of dronabinol during detoxification and the initiation of treatment with extended release naltrexone. *Drug Alcohol Depend*, 154(1), 38–45. <https://doi.org/10.1016/j.drugalcdep.2015.05.013>
- Boks, P., Joober, R., & Kirmayer, L. J. (2015). Mental wellness in Canada's Aboriginal communities: Striving toward reconciliation. *J Psychiatry Neurosci*, 40(6), 363–365. <https://doi.org/10.1503/jpn.150309>
- Bonn-Miller, M. O., Boden, M. T., Bucossi, M. M., & Babson, K. A. (2014). Self-reported cannabis use characteristics, patterns and helpfulness among medical cannabis users. *The American Journal of Drug and Alcohol Abuse*, 40(1), 23–30. <https://doi.org/10.3109/00952990.2013.821477>
- Bowleg, L. (2012). The problem with the phrase women and minorities: Intersectionality-an important theoretical framework for public health. *American Journal of Public Health*, 102(7), 1267–1273. <https://doi.org/10.2105/AJPH.2012.300750>
- Bradby, H., Varyani, M., Oglethorpe, R., Raine, W., White, I., & Helen, M. (2007). British Asian families and the use of child and adolescent mental health services: A qualitative study of a hard to reach group. *Soc Sci Med*, 65(12), 2413–2424. <https://doi.org/10.1016/j.socscimed.2007.07.025>
- Braun, V., & Clarke, V. (2006). Using thematic analysis in psychology. *Qualitative Research in Psychology*, 3(2), 77–101. <https://doi.org/10.1191/1478088706qp063oa>
- Buckner, J. D., Shah, S. M., Dean, K. E., & Zvolensky, M. J. (2016). Cannabis use frequency and use-related impairment among African-American and white users: The impact of cannabis use motives. *Ethnicity & Health*, 21(3), 318–331. <https://doi.org/10.1080/13557858.2015.1065311>
- Carstairs, C. (2006). *Jailed for possession: Illegal drug use, regulation, and power in Canada, 1920–1961*. University of Toronto Press.
- Chen, A. W., & Kazanjian, A. (2005). Rate of mental health service utilization by Chinese immigrants in British Columbia. *Canadian Journal of Public Health*, 96(1), 49–51. <https://doi.org/10.1007/BF03404017>
- Chou, F. Y., Armstrong, H. L., Wang, L., Bacani, N., Lachowsky, N. J., Patterson, T. L., Walsh, Z., Olarewaju, G., Card, K. G., Roth, E. A., Hogg, R. S., & Moore, D. M. (2019). A longitudinal analysis of cannabis use and mental health symptoms among gay, bisexual, and other men who have sex with men in Vancouver, Canada. *Journal of Affective Disorders*, 247, 125–133. <https://doi.org/10.1016/j.jad.2019.01.015>
- Clement, F., Corbett, C., Farkas, B., Diaz, R., Taplin, J., Hill, M., Patten, S., & Haines-Saah, R. (2019). *Cannabis and mental health: Priorities for research*. https://mentalhealthcommission.ca/wp-content/uploads/2021/09/Cannabis_mental_Health_Summary_july_2019_eng.pdf
- Crippa, J. A., Zuardi, A. W., Martin-Santos, R., Bhattacharyya, S., Atakan, Z., McGuire, P., & Fusar-Poli, P. (2009). Cannabis and anxiety: A critical review of the evidence. *Human Psychopharmacology*, 24(7), 515–523. <https://doi.org/10.1002/hup.1048>
- Das, A., Hendershot, C. S., Husain, M. I., Knyahnytska, Y., Elsaid, S., Le Foll, B., & Kloiber, S. (2024). Perceptions, experiences, and patterns of Cannabis use in individuals with mood and anxiety disorders in the

- context of Cannabis legalization and medical Cannabis program in Canada - A qualitative study. *Pharmacopsychiatry*, 57(3), 141–151. <https://doi.org/10.1055/a-2264-1047>
- Day, N. L., Goldschmidt, L., Day, R., Larkby, C., & Richardson, G. A. (2015). Prenatal marijuana exposure, age of marijuana initiation, and the development of psychotic symptoms in young adults. *Psychological Medicine*, 45(8), 1779–1787. <https://doi.org/10.1017/S0033291714002906>
- Degenhardt, L., Hall, W., & Lynskey, M. (2003). Exploring the association between cannabis use and depression. *Addiction*, 98(11), 1493–1504. <https://doi.org/10.1046/j.1360-0443.2003.00437.x>
- DeSon, J. J., & Andover, M. S. (2024). Microaggressions toward sexual and gender minority emerging adults: An updated systematic review of psychological correlates and outcomes and the role of intersectionality. *LGBT Health*, 11(4), 249–268. <https://doi.org/10.1089/lgbt.2023.0032>
- Dyar, C., Newcomb, M. E., & Mustanski, B. (2019). Longitudinal associations between minority stressors and substance use among sexual and gender minority individuals. *Drug Alcohol Dependence*, 201, 205–211. <https://doi.org/10.1016/j.drugalcdep.2019.03.032>
- Dyar, C., Sarno, E. L., Newcomb, M. E., & Whitton, S. W. (2020). Longitudinal associations between minority stress, internalizing symptoms, and substance use among sexual and gender minority individuals assigned female at birth. *Journal of Consulting and Clinical Psychology*, 88(5), 389–401. <https://doi.org/10.1037/ccp0000487>
- Giffen, P. J., Endicott, S. J., & Boorman, S. (1991). Panic and indifference: the politics of Canada's drug laws : a study in the sociology of law. Canadian Centre on Substance Abuse.
- Gobbi, G., Atkin, T., Zytynski, T., Wang, S., Askari, S., Boruff, J., Ware, M., Marmorstein, N., Cipriani, A., Dendukuri, N., & Mayo, N. (2019). Association of Cannabis use in adolescence and risk of depression, anxiety, and suicidality in young adulthood: A systematic review and meta-analysis. *JAMA Psychiatry*, 76(4), 426–434. <https://doi.org/10.1001/jamapsychiatry.2018.4500>
- Government of Canada. (2019). *Canadian Alcohol and Drugs Survey (CADS): summary of results for 2019*. <https://www.canada.ca/en/health-canada/services/canadian-alcohol-drugs-survey/2019-summary.html>
- Government of Canada. (2021). *Cannabis Legalization and Regulation*. <https://www.justice.gc.ca/eng/cj-jp/cannabis/>
- Government of Canada. (2022). *Summary of results for the Canadian Student Tobacco, Alcohol and Drugs Survey 2021–22*. <https://www.canada.ca/en/health-canada/services/canadian-student-tobacco-alcohol-drugs-survey/2021-2022-summary.html>
- Government of Canada. (2023). *Canadian Cannabis Survey 2023: Summary*. <https://www.canada.ca/en/health-canada/services/drugs-medication/cannabis/research-data/canadian-cannabis-survey-2023-summary.html>
- Government of Ontario. (2024). *Data Standards for the Identification and Monitoring of Systemic Racism*. King's Printer for Ontario. <https://www.ontario.ca/document/data-standards-identification-and-monitoring-systemic-racism>
- Han, B. H., Yang, K. H., Cleland, C. M., & Palamar, J. J. (2025). Trends in past-month cannabis use among older adults. *JAMA Internal Medicine*, 185(7), 881–883. <https://doi.org/10.1001/jamainternmed.2025.1156>
- Harrell, J. P., Hall, S., & Taliaferro, J. (2003). Physiological responses to racism and discrimination: An assessment of the evidence. *American Journal of Public Health*, 93(2), 243–248. <https://doi.org/10.2105/AJPH.93.2.243>
- Heard, E., Fitzgerald, L., Wigginton, B., & Mutch, A. (2020). Applying intersectionality theory in health promotion research and practice. *Health Promotion International*, 35(4), 866–876. <https://doi.org/10.1093/heapro/daz080>
- Henquet, C., Murray, R., Linszen, D., & van Os, J. (2005). The environment and schizophrenia: The role of cannabis use. *Schizophrenia Bulletin*, 31(3), 608–612. <https://doi.org/10.1093/schbul/sbi027>
- Hoch, E., Niemann, D., von Keller, R., Schneider, M., Friemel, C. M., Preuss, U. W., Hasan, A., & Pogarell, O. (2019). How effective and safe is medical cannabis as a treatment of mental disorders? A systematic review. *European Archives of Psychiatry and Clinical Neuroscience*, 269(1), 87–105. <https://doi.org/10.1007/s00406-019-00984-4>
- Hunt, G. E., Large, M. M., Cleary, M., Lai, H. M. X., & Saunders, J. B. (2018). Prevalence of comorbid substance use in schizophrenia spectrum disorders in community and clinical settings, 1990–2017: Systematic review and meta-analysis. *Drug Alcohol Dependence*, 191, 234–258. <https://doi.org/10.1016/j.drugalcdep.2018.07.011>

- IBM Corp. (2017). *IBM SPSS statistics for windows, version 25.0*.
- Jeffers, A. M., Glantz, S., Byers, A., & Keyhani, S. (2021). Sociodemographic characteristics associated with and prevalence and frequency of Cannabis use among adults in the US. *JAMA Network Open*, 4(11), e2136571. <https://doi.org/10.1001/jamanetworkopen.2021.36571>
- Jetly, R., Heber, A., Fraser, G., & Boisvert, D. (2015). The efficacy of nabilone, a synthetic cannabinoid, in the treatment of PTSD-associated nightmares: A preliminary randomized, double-blind, placebo-controlled cross-over design study. *Psychoneuroendocrinology*, 51, 585–588. <https://doi.org/10.1016/j.psyneuen.2014.11.002>
- Kafele, K. (2004). *Racial discrimination and mental health in racialized and Aboriginal communities*. . <https://www3.ohrc.on.ca/en/race-policy-dialogue-papers/racial-discrimination-and-mental-health-racialized-and-aboriginal-communities>.
- Khan, S. (2017). *Concurrent mental and substance use disorders in Canada, in Health Reports*.
- Khenti, A. (2014). The Canadian war on drugs: Structural violence and unequal treatment of Black Canadians. *International Journal of Drug Policy*, 25(2), 190–195. <https://doi.org/10.1016/j.drugpo.2013.12.001>
- Kiburi, S. K., Molebatsi, K., Ntlantsana, V., & Lynskey, M. T. (2021). Cannabis use in adolescence and risk of psychosis: Are there factors that moderate this relationship? A systematic review and meta-analysis. *Substance Abuse*, 42(4), 527–542. <https://doi.org/10.1080/08897077.2021.1876200>
- Konefal, S., Gabrys, R., & Porath, A. J. (2019). *Clearing the smoke on cannabis: Regular use and mental health*.
- Leong, F. T., & Lau, A. S. (2001). Barriers to providing effective mental health services to Asian Americans. *Mental Health Services Research*, 3(4), 201–214. <https://doi.org/10.1023/a:1013177014788>
- Lev-Ran, S., Roerecke, M., Le Foll, B., George, T. P., McKenzie, K., & Rehm, J. (2014). The association between cannabis use and depression: A systematic review and meta-analysis of longitudinal studies. *Psychological Medicine*, 44(4), 797–810. <https://doi.org/10.1017/S0033291713001438>
- Loya, F., Reddy, R., & Hinshaw, S. P. (2010). Mental illness stigma as a mediator of differences in Caucasian and south Asian college students' attitudes toward psychological counseling. *Journal of Counseling Psychology*, 57(4), 484–490. <https://doi.org/10.1037/a0021113>
- Ltd, Q. I. P. (2020). NVivo. In.
- Macgregor, C., Walumbe, J., Tulle, E., Seenan, C., & Blane, D. N. (2023). Intersectionality as a theoretical framework for researching health inequities in chronic pain. *British Journal of Pain*, 17(5), 479–490. <https://doi.org/10.1177/20494637231188583>
- Mammen, G., Rueda, S., Roerecke, M., Bonato, S., Lev-Ran, S., & Rehm, J. (2018). Association of Cannabis with long-term clinical symptoms in anxiety and mood disorders: A systematic review of prospective studies. *Journal of Clinical Psychiatry*, 79(4), 17r11839. <https://doi.org/10.4088/JCP.17r11839>
- Marshall, S. G. (2015). Canadian Drug policy and the reproduction of Indigenous inequities. *International Indigenous Policy Journal*, 6(1), 7. <https://doi.org/10.18584/iipj.2015.6.1.7>
- McLoughlin, B. C., Pushpa-Rajah, J. A., Gillies, D., Rathbone, J., Variend, H., Kalakouti, E., & Kyprianou, K. (2014). Cannabis and schizophrenia. *Cochrane Database of Systematic Reviews*, 2014(10), CD004837. <https://doi.org/10.1002/14651858.CD004837.pub3>
- Mosher, C., & Akins, S. (2015). *Drugs and drug control in Canada. Pan African issues in drugs and drug control-An international perspective, in Pan-African issues in drugs and drug control: An international perspective*. Routledge.
- Mukhopadhyay, B., Thambinathan, V., & Kinsella, E. A. (2024). Towards anti-racist futures: A scoping review exploring educational interventions that address systemic racism in post graduate medical education. *Advances in Health Sciences Education: Theory and Practice*, 30(2), 359–381. <https://doi.org/10.1007/s10459-024-10343-1>
- Ng, C. H. (1997). The stigma of mental illness in Asian cultures. *Australian & New Zealand Journal of Psychiatry*, 31(3), 382–390. <https://doi.org/10.3109/00048679709073848>
- Owusu-Bempah, A., & Gabbidon, S. (2020). *Race, Ethnicity, Crime, and Justice: An International Dilemma* (2nd Ed.). Routledge.

- Owusu-Bempah, A., & Rehmatullah, T. (2023). *Waiting to Exhale: Cannabis Legalization and the Fight for Racial Justice*. The MIT Press.
- Scheim, A. I., Bauer, G. R., & Shokoohi, M. (2017). Drug use among transgender people in Ontario, Canada: Disparities and associations with social exclusion. *Addictive Behaviors*, 72, 151–158. <https://doi.org/10.1016/j.addbeh.2017.03.022>
- Schubart, C. D., van Gastel, W. A., Breetvelt, E. J., Beetz, S. L., Ophoff, R. A., Sommer, I. E., Kahn, R. S., & Boks, M. P. (2011). Cannabis use at a young age is associated with psychotic experiences. *Psychological Medicine*, 41(6), 1301–1310. <https://doi.org/10.1017/S003329171000187X>
- Seng, J. S., Lopez, W. D., Sperlich, M., Hamama, L., & Reed Meldrum, C. D. (2012). Marginalized identities, discrimination burden, and mental health: Empirical exploration of an interpersonal-level approach to modeling intersectionality. *Social Science & Medicine*, 75(12), 2437–2445. <https://doi.org/10.1016/j.socscimed.2012.09.023>
- Sentell, T., Zhang, W., Davis, J., Baker, K. K., & Braun, K. L. (2014). The influence of community and individual health literacy on self-reported health status. *Journal of General Internal Medicine*, 29(2), 298–304. <https://doi.org/10.1007/s11606-013-2638-3>
- Smolkina, M., Morley, K. I., Rijdsdijk, F., Agrawal, A., Bergin, J. E., Nelson, E. C., Statham, D., Martin, N. G., & Lynskey, M. T. (2017). Cannabis and depression: A twin model approach to co-morbidity. *Behavior Genetics*, 47(4), 394–404. <https://doi.org/10.1007/s10519-017-9848-0>
- Sockalingam, S., Arena, A., Serhal, E., Mohri, L., Alloo, J., & Crawford, A. (2018). Building provincial mental health capacity in primary care: An evaluation of a project ECHO mental health program. *Academic Psychiatry*, 42(4), 451–457. <https://doi.org/10.1007/s40596-017-0735-z>
- Souissi, T. (2022). Racialized Minorities. In *The Canadian Encyclopedia*. <https://www.thecanadianencyclopedia.ca/en/article/racialized-minorities>.
- Stuart-Maver, S. L. (2020). Working with clients who self-medicate using cannabis: Ethical and clinical considerations for psychologists. *Professional Psychology: Research and Practice*, 51(1), 77–84. <https://doi.org/10.1037/pro0000269>
- Tanovich, D. M. (2006). The further erasure of race in charter cases. 8 *Criminal Reports*, 6, 38–44. Available at SSRN: <https://ssrn.com/abstract=2741245>
- Turna, J., Balodis, I., Munn, C., Van Ameringen, M., Busse, J., & MacKillop, J. (2020). Overlapping patterns of recreational and medical cannabis use in a large community sample of cannabis users. *Comprehensive Psychiatry*, 102, 152188. <https://doi.org/10.1016/j.comppsy.2020.152188>
- Utsey, S. O., Giesbrecht, M., Hook, J., & Stanard, P. M. (2008). Cultural, sociofamilial, and psychological resources that inhibit psychological distress in African Americans exposed to stressful life events and race-related stress. *Journal of Counseling Psychology*, 55(1), 49–62. <https://doi.org/10.1037/0022-0167.55.1.49>
- Weinberger, A. H., Pacek, L. R., Sheffer, C. E., Budney, A. J., Lee, J., & Goodwin, R. D. (2019). Serious psychological distress and daily cannabis use, 2008 to 2016: Potential implications for mental health? *Drug Alcohol Dependence*, 197, 134–140. <https://doi.org/10.1016/j.drugalcdep.2019.01.010>
- Winter, S., Diamond, M., Green, J., Karasic, D., Reed, T., Whittle, S., & Wylie, K. (2016). Transgender people: Health at the margins of society. *Lancet*, 388(10042), 390–400. [https://doi.org/10.1016/S0140-6736\(16\)00683-8](https://doi.org/10.1016/S0140-6736(16)00683-8)
- Wortley, S., & Tanner, J. (2003). Data, denials, and confusion: The racial profiling debate in Toronto. *Canadian Journal of Criminology and Criminal Justice*, 45(3), 367–389. <https://doi.org/10.3138/cjccj.45.3.367>
- Zhou, C., Crawford, A., Serhal, E., Kurdyak, P., & Sockalingam, S. (2016). The impact of project ECHO on participant and patient outcomes: A systematic review. *Academic Medicine*, 91(10), 1439–1461. <https://doi.org/10.1097/ACM.0000000000001328>